

PERINGATAN!

Dokumentasi ini berisi simulasi serangan siber yang dilakukan di dalam lingkungan **laboratorium yang terkontrol** terhadap target yang **sah/legal**.

Melakukan serangan siber di lingkungan **produksi (nyata)** atau terhadap target yang **tidak sah** merupakan tindakan ilegal yang **melanggar hukum**.

Penulis **tidak bertanggung jawab** atas segala efek penyalahgunaan informasi dalam dokumentasi ini untuk kegiatan-kegiatan ilegal.

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Profil Penyerang

Parameter	Detail
Nama Sistem Operasi	Kali Linux
Asal Sistem Operasi	Linux
Versi Sistem Operasi	2026.2
Dasar Sistem Operasi	Debian
Versi Dasar Sistem Operasi	Debian 10.x 64-bit
Bentuk Sistem Operasi	Mesin Virtual (VM)
Platform Virtualisasi	VMware Workstation Pro

Profil Target

Parameter	Detail
Nama Sistem Operasi	Metasploitable 2
Asal Sistem Operasi	Linux
Versi Sistem Operasi	2.6.X
Dasar Sistem Operasi	Ubuntu
Versi Dasar Sistem Operasi	Ubuntu 8.04 LTS
Bentuk Sistem Operasi	Mesin Virtual (VM)
Platform Virtualisasi	VMware Workstation Pro

Profil Laboratorium

Parameter	Detail
Nama Laboratorium	VMware Workstation Pro

Fase 1: Reconnaissance

Reconnaissance (pengintaian) adalah fase pertama dalam rantai serangan siber (*Cyber Attack Chain / Cyber Kill Chain*). Pada fase ini, penyerang (*attacker*) mengidentifikasi target, memastikan target aktif, memetakan pintu masuk (*port*), mengidentifikasi layanan (*service*) yang berjalan pada target, serta mencari celah keamanan atau kerentanan (*vulnerability*) yang dimiliki oleh target. Data yang dikumpulkan pada fase ini akan digunakan untuk merancang serangan (*weaponization*) dan mengeksploitasi kerentanan target pada tahap berikutnya.

1.1. Menemukan Target

Perintah: ip a

`ip a` merupakan singkatan dari `ip addr` atau `ip address`. Perintah ini digunakan untuk menampilkan dan mengonfigurasi alamat IP serta antarmuka jaringan pada sistem operasi tempat perintah tersebut dijalankan. Dalam dokumentasi ini, `ip a` digunakan untuk mengetahui alamat IP mesin target secara langsung guna mempercepat proses praktikum. Di dunia nyata, administrator target dapat menggunakan perintah ini, namun informasi alamat IP internal biasanya bersifat rahasia dan tidak dipublikasikan ke publik. Perintah ini dijalankan pada mesin penyerang maupun target untuk verifikasi.

- Alamat IP Kali Linux: 192.168.100.131/24
- Alamat IP Metasploitable 2: 192.168.100.132/24

Perintah: netdiscover (sudo)

`netdiscover` adalah alat pemindaian jaringan aktif/pasif yang digunakan untuk menemukan perangkat yang terhubung dalam satu segmen jaringan. Perintah ini memanfaatkan protokol ARP (*Address Resolution Protocol*) untuk mengidentifikasi perangkat dengan mengumpulkan informasi berupa alamat IP, alamat MAC, jumlah paket ARP yang terkirim, serta vendor kartu jaringan

(*Network Interface Card / NIC*) atau *hostname*. Perintah ini dijalankan dari mesin Kali Linux untuk mendeteksi keberadaan mesin Metasploitable 2.

- Segmen Jaringan: 192.168.100.x/24
- Alamat IP 192.168.100.1 adalah milik sistem operasi *host* yang menjalankan VMware.
- Alamat IP 192.168.100.2 adalah milik *virtual gateway/router* internal VMware.
- Alamat IP 192.168.100.132 adalah milik mesin target (Metasploitable 2).
- Alamat IP 192.168.100.254 adalah milik layanan DHCP Server internal VMware.
- Alamat IP 192.168.100.255 adalah alamat *broadcast* jaringan VMware.

Perintah: arp-scan (sudo)

arp-scan adalah alat pemindaian baris perintah (*command-line*) yang mengirimkan paket ARP Request ke setiap alamat IP pada segmen jaringan yang ditentukan secara berurutan. Seperti netdiscover, alat ini memetakan hubungan antara alamat IP dan alamat MAC di jaringan lokal guna menemukan perangkat aktif secara cepat.

- Segmen Jaringan: 192.168.100.x/24
Interface: eth0 (ethernet0)
Type: EN10MB
- Alamat IP 192.168.100.1 adalah milik sistem operasi *host* yang menjalankan VMware.
- Alamat IP 192.168.100.2 adalah milik *virtual gateway/router* internal VMware.
- Alamat IP 192.168.100.132 adalah milik mesin target (Metasploitable 2).
- Alamat IP 192.168.100.254 adalah milik layanan DHCP Server internal VMware.

- Alamat IP 192.168.100.255 adalah alamat *broadcast* jaringan VMware.

Perintah: ping

Utilitas ping digunakan untuk memverifikasi apakah host tujuan dalam keadaan aktif dan dapat dijangkau melalui jaringan. Utilitas ini mengirimkan paket ICMP (*Internet Control Message Protocol*) *Echo Request* ke alamat target dan menunggu respons berupa *Echo Reply*. Jika target merespons, berarti koneksi terhubung. Jika tidak ada balasan (*Request Timed Out* atau *Destination Host Unreachable*), target kemungkinan dalam keadaan tidak aktif, memblokir paket ICMP (misalnya karena aturan firewall), atau terjadi gangguan pada rute jaringan.

1.2. Memetakan Port dan Layanan Aktif

Perintah: nmap

nmap (*Network Mapper*) adalah perangkat lunak pemindaian keamanan jaringan yang digunakan untuk mengidentifikasi host dan layanan aktif di jaringan komputer. Utilitas ini bekerja dengan memindai port pada perangkat target. Total port yang tersedia pada sebuah antarmuka jaringan adalah 65.535 port, yang terbagi menjadi tiga kategori:

1. **Well-Known Ports (0 - 1023):** Dicadangkan untuk layanan jaringan standar dan protokol inti sistem (misalnya HTTP, FTP, SSH).
2. **Registered Ports (1024 - 49.151):** Digunakan untuk aplikasi atau perangkat lunak pihak ketiga tertentu.
3. **Dynamic / Private Ports (49.152 - 65.535):** Digunakan untuk koneksi sementara secara dinamis.

Selain mendeteksi status port (terbuka, tertutup, atau terfilter), nmap juga dapat mengidentifikasi jenis layanan beserta versinya (*service/version detection*) serta mendeteksi sistem operasi target.

Berikut adalah daftar port yang berhasil dipetakan nmap:

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp	open	exec	Netkit-rsh rexecd
513/tcp	open	login	OpenBSD or Solaris rlogind
514/tcp	open	tcpwrapped	-
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5

PORT	STATE	SERVICE	VERSION
3632/tcp	open	distccd	distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	(access denied)
6667/tcp	open	irc	UnrealIRCd
6697/tcp	open	irc	UnrealIRCd
8009/tcp	open	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1
8787/tcp	open	drb	Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
38057/tcp	open	status	1 (RPC #100024)
42189/tcp	open	nlockmgr	1-4 (RPC #100021)
57819/tcp	open	mountd	1-3 (RPC #100005)
60910/tcp	open	java-rmi	GNU Classpath grmiregistry

Tabel 1.2.1 Daftar port hasil pemindaian Nmap

1.3. Menemukan Kerentanan Web

Perintah: nikto

nikto adalah pemindai kerentanan server web (*web server scanner*) sumber terbuka (open-source) yang dirancang untuk mendeteksi konfigurasi yang tidak aman, file default/sensitif, serta kerentanan umum pada layanan web (HTTP port 80 atau HTTPS port 443). Pengujian ini dijalankan dari mesin Kali Linux terhadap layanan HTTP target.

- **Ringkasan Temuan Kerentanan Web**

Berdasarkan hasil pemindaian di atas, berikut adalah beberapa temuan penting dan kerentanan kritis pada server web target:

No	Kategori Temuan	Detail Temuan	Potensi Risiko / Dampak
1	Outdated Server Software	Menggunakan Apache/2.2.8 dan PHP/5.2.4 (versi usang).	Rentan terhadap eksploitasi celah keamanan lama yang sudah diketahui publik (<i>known vulnerabilities</i>).
2	HTTP Methods Allowed	Metode HTTP TRACE aktif pada server.	Rentan terhadap serangan <i>Cross-Site Tracing</i> (XST) untuk mencuri cookie sesi.
3	Vulnerable Directories	Direktori /dav/ (WebDAV) aktif dengan konfigurasi lemah.	Memungkinkan penyerang mengunggah berkas jahat (<i>web shell</i>) ke server menggunakan metode HTTP PUT.
4	Exposed Interfaces	Ditemukan halaman login phpMyAdmin, wiki (/twiki/), dan utilitas administrasi lainnya.	Rentan terhadap serangan <i>brute force</i> pada kredensial admin atau eksploitasi langsung pada aplikasi web.

Tabel 1.3.1 Daftar port hasil pemindaian nikto

1.4. Menemukan File Tersembunyi dengan URL Web

Perintah: **dirb** (DirBuster)

dirb (Directory Buster) adalah utilitas baris perintah yang digunakan untuk mencari direktori dan file tersembunyi pada sebuah situs web. Alat ini bekerja dengan metode *brute-force* berbasis kamus kata (*wordlist*). Jika direktori atau file dalam daftar ditemukan pada server web target, dirb akan melaporkan kode respons HTTP yang sesuai (seperti 200 OK atau 403 Forbidden). Beberapa direktori yang memiliki konfigurasi *directory listing* (dapat diakses/dilihat isinya secara langsung) juga akan terdeteksi. Pengujian ini dijalankan dari mesin Kali Linux.

- **Ringkasan Direktori Kritis yang Ditemukan**

Dari hasil pemindaian menggunakan dirb, ditemukan beberapa direktori penting yang menjadi pintu masuk utama untuk eksploitasi:

Direktori	Kode HTTP	Keterangan / Fungsi	Signifikansi Eksploitasi
/dvwa/	302 / 200	Damn Vulnerable Web Application.	Aplikasi web rentan untuk latihan penetrasi; memiliki celah SQL injection, XSS, file inclusion, dll.
/mutillidae/	302 / 200	OWASP Mutillidae II.	Sistem web rentan yang dikembangkan khusus untuk latihan keamanan siber berbasis OWASP Top 10.
/phpMyAdmin/	302 / 200	Halaman administrasi MySQL.	Akses ke basis data server jika penyerang berhasil menebak kredensial atau

Direktori	Kode HTTP	Keterangan / Fungsi	Signifikansi Eksploitasi
			mengeksplorasi celah di phpMyAdmin.
/twiki/	200	Kolaborasi wiki TWiki.	Platform wiki yang rentan terhadap kerentanan Remote Code Execution (RCE) versi lama.
/dav/	200	Direktori WebDAV.	Folder yang dapat digunakan untuk menyimpan dan mengeksekusi payload karena metode tulis (write/PUT) diaktifkan.
/security /	200	Pengaturan keamanan Metasploitable.	Berisi informasi konfigurasi tingkat keamanan situs target.

Tabel 1.4.1 Daftar port hasil pemindaian dirb (dirbuster)

LAMPIRAN

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:fb:f8:f9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.131/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
        valid_lft 1315sec preferred_lft 1315sec
    inet6 fe80::4e73:669a:ff63:f2e5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Gambar 1, Command ip a pada Kali Linux

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:ec:92:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.132/24 brd 192.168.100.255 scope global eth0
    inet6 fe80::20c:29ff:feec:9223/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 00:0c:29:ec:92:2d brd ff:ff:ff:ff:ff:ff
```

Gambar 2, Command ip a pada Metasploitable 2

```
Currently scanning: Finished! | Screen View: Unique Hosts

4 Captured ARP Req/Rep packets, from 4 hosts. Total size: 240

  IP             At MAC Address      Count  Len  MAC Vendor / Hostname
-----
192.168.100.1    00:50:56:c0:00:08    1      60  VMware, Inc.
192.168.100.2    00:50:56:e4:f8:8d    1      60  VMware, Inc.
192.168.100.132  00:0c:29:ec:92:23    1      60  VMware, Inc.
192.168.100.254  00:50:56:ed:10:09    1      60  VMware, Inc.
```

Gambar 3, Command netdiscover pada Kali Linux

```
(kali@kali)-[~]
$ sudo arp-scan --localnet
Interface: eth0, type: EN10MB, MAC: 00:0c:29:fb:f8:f9, IPv4: 192.168.100.131
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.100.1    00:50:56:c0:00:08    (Unknown)
192.168.100.2    00:50:56:e4:f8:8d    (Unknown)
192.168.100.132  00:0c:29:ec:92:23    (Unknown)
192.168.100.254  00:50:56:ed:10:09    (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.876 seconds (136.46 hosts/sec). 4 responded
```

Gambar 4, Command arp-scan pada Kali Linux

```

(kali㉿kali)-[~]
$ ping 192.168.100.132
PING 192.168.100.132 (192.168.100.132) 56(84) bytes of data.
 64 bytes from 192.168.100.132: icmp_seq=1 ttl=64 time=1.80 ms
 64 bytes from 192.168.100.132: icmp_seq=2 ttl=64 time=0.698 ms
 64 bytes from 192.168.100.132: icmp_seq=3 ttl=64 time=0.636 ms
 64 bytes from 192.168.100.132: icmp_seq=4 ttl=64 time=0.724 ms
 64 bytes from 192.168.100.132: icmp_seq=5 ttl=64 time=0.842 ms
^C
— 192.168.100.132 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4066ms
rtt min/avg/max/mdev = 0.636/0.940/1.804/0.436 ms

```

Gambar 5, Kali Linux mencoba ping menuju Metasploitable 2

```

(kali㉿kali)-[~]
$ nmap 192.168.100.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 09:31 -0400
Nmap scan report for 192.168.100.132
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:EC:92:23 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

```

Gambar 6, Kali menjalankan nmap pada alamat IP Metasploitable 2

```

(kali@kali)-[~]
$ nmap -A -p- 192.168.100.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 09:33 -0400
Nmap scan report for 192.168.100.132
Host is up (0.0014s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.100.131
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-07-25T13:35:49+00:00; +1s from scanner time.
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_RC4_128_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN

```

Gambar 7, Kali menjalankan nmap pada alamat IP Metasploitable 2

```

53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100000 2 111/tcp rpcbind
|_100000 2 111/udp rpcbind
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/udp nfs
|_100005 1,2,3 38140/udp mountd
|_100005 1,2,3 57819/tcp mountd
|_100021 1,3,4 42189/tcp nlockmgr
|_100021 1,3,4 58652/udp nlockmgr
|_100024 1 38057/tcp status
|_100024 1 48809/udp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|_Protocol: 10
|_Version: 5.0.51a-3ubuntu5
|_Thread ID: 8
|_Capabilities flags: 43564
|_Some Capabilities: ConnectWithDatabase, Support41Auth, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, SupportsCompression, LongColumnFlag
|_Status: Autocommit
|_Salt: MH`L`Bgb4|9AzdfRGP
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))

```

Gambar 8, Kali menjalankan nmap pada alamat IP Metasploitable 2

```

5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-07-25T13:35:49+00:00; +1s from scanner time.
5900/tcp open  vnc VNC (protocol 3.3)
|_vnc-info:
|_Protocol version: 3.3
|_Security types:
|_VNC Authentication (2)
6000/tcp open  X11 (access denied)
6667/tcp open  irc UnrealIRCd
|_irc-info:
|_users: 1
|_servers: 1
|_lusers: 1
|_lservers: 0
|_server: irc.Metasploitable.LAN
|_version: Unreal3.2.8.1. irc.Metasploitable.LAN
|_uptime: 0 days, 0:22:15
|_source ident: nmap
|_source host: 5F1669D2.2ADA0512.FFFA6D49.IP
|_error: Closing Link: jsumezyoy[192.168.100.131] (Quit: jsumezyoy)
6697/tcp open  irc UnrealIRCd
8009/tcp open  ajp13 Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
8787/tcp open  drb Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
38057/tcp open status 1 (RPC #100024)
42189/tcp open nlockmgr 1-4 (RPC #100021)
57819/tcp open mountd 1-3 (RPC #100005)
60910/tcp open java-rmi GNU Classpath grmiregistry
MAC Address: 00:0C:29:EC:92:23 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

Gambar 9, Kali menjalankan nmap pada alamat IP Metasploitable 2

```

Host script results:
|_ smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-07-25T09:35:41-04:00
|_ smb-security-mode:
|_ account_used: guest
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s

TRACEROUTE
HOP RTT ADDRESS
1 1.43 ms 192.168.100.132

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 144.61 seconds

```

Gambar 10, Kali menjalankan nmap pada alamat IP Metasploitable 2

```

(kali@kali)-[~]
$ nikto -h http://192.168.100.132
- Nikto v2.6.0

+ Target IP: 192.168.100.132
+ Target Hostname: 192.168.100.132
+ Target Port: 80
+ Platform: Unknown
+ Start Time: 2026-07-25 09:42:17 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ ERROR: Failed to check for updates: 403
+ [999986] /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ [750500] /icons/: Directory indexing found.
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [999100] /index: Uncommon header(s) 'tcn' found, with contents: list.
+ [999965] /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives fo
it/sectou.php?id=4698ebdc59d15,https://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ [600625] PHP/5.2.4-2ubuntu5.10 appears to be outdated (current is at least 8.5.1).
+ [600050] Apache/2.2.8 appears to be outdated (current is at least 2.4.66).
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Optio
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [999967] /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ [000434] /: HTTP TRACE method is active and replies which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_
+ [750510] /phpinfo.php: Output from the phpinfo() function was found.
+ [750500] /doc/: Directory indexing found.
+ [001213] /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ [001384] /?PHPBB85F2A0-3C92-11d0-A3A9-4C7B08C10000: PHP Easter Eggs reveals potentially sensitive information via HTTP requests that contain specific QUE
-you-dare-to-show-your-php-easter-egg/
+ [001385] /?PHPBB85F2A0-3C92-11d0-A3A9-4C7B08C10000: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUE
you-dare-to-show-your-php-easter-egg/
+ [001386] /?PHPBB85F2A0-3C92-11d0-A3A9-4C7B08C10000: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUE
you-dare-to-show-your-php-easter-egg/
+ [001387] /?PHPBB85F2A0-3C92-11d0-A3A9-4C7B08C10000: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUE
you-dare-to-show-your-php-easter-egg/
+ [001795] /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [999984] /phpMyAdmin/ChangeLog: Server may leak inodes via ETags, header found with file /phpMyAdmin/ChangeLog, inode: 92462, size: 40540, mtime: Tue Dec
ename.cgi?name=CVE-2003-1418
+ [001796] /phpMyAdmin/ChangeLog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.

```

Gambar 11, Kali menjalankan nikto pada protokol HTTP Metasploitable 2

```

+ [001796] /phpMyAdmin/ChangeLog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [750500] /test/: Directory indexing found.
+ [001894] /test/: This might be interesting.
+ [002989] /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ [003584] /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/

```

Gambar 12, Kali menjalankan nmap pada protokol HTTP Metasploitable 2

```

(kali@kali)-[~]
$ dirb http://192.168.100.132

DIRB v2.22
By The Dark Raver

START_TIME: Sat Jul 25 09:45:13 2026
URL_BASE: http://192.168.100.132/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

--- Scanning URL: http://192.168.100.132/ ---
+ http://192.168.100.132/cgi-bin/ (CODE:403|SIZE:296)
=> DIRECTORY: http://192.168.100.132/dav/
+ http://192.168.100.132/index (CODE:200|SIZE:891)
+ http://192.168.100.132/index.php (CODE:200|SIZE:891)
+ http://192.168.100.132/phpinfo (CODE:200|SIZE:48107)
+ http://192.168.100.132/phpinfo.php (CODE:200|SIZE:48119)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/
+ http://192.168.100.132/server-status (CODE:403|SIZE:301)
=> DIRECTORY: http://192.168.100.132/test/
=> DIRECTORY: http://192.168.100.132/twiki/

--- Entering directory: http://192.168.100.132/dav/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/ ---
+ http://192.168.100.132/phpMyAdmin/calendar (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.100.132/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/contrib/
+ http://192.168.100.132/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.100.132/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.100.132/phpMyAdmin/import (CODE:200|SIZE:4145)

```

Gambar 13, Kali menjalankan dirb pada protokol HTTP Metasploitable 2

```

+ http://192.168.100.132/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.100.132/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/contrib/
+ http://192.168.100.132/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.100.132/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.100.132/phpMyAdmin/import (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/index (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/index.php (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/js/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/lang/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/libraries/
+ http://192.168.100.132/phpMyAdmin/license (CODE:200|SIZE:18011)
+ http://192.168.100.132/phpMyAdmin/LICENSE (CODE:200|SIZE:18011)
+ http://192.168.100.132/phpMyAdmin/main (CODE:200|SIZE:4227)
+ http://192.168.100.132/phpMyAdmin/navigation (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/phpinfo (CODE:200|SIZE:0)
+ http://192.168.100.132/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
+ http://192.168.100.132/phpMyAdmin/phpmyadmin (CODE:200|SIZE:21389)
+ http://192.168.100.132/phpMyAdmin/print (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/readme (CODE:200|SIZE:2624)
+ http://192.168.100.132/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://192.168.100.132/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://192.168.100.132/phpMyAdmin/robots.txt (CODE:200|SIZE:26)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/scripts/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/
+ http://192.168.100.132/phpMyAdmin/sql (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/test/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/themes/
+ http://192.168.100.132/phpMyAdmin/TODO (CODE:200|SIZE:235)
+ http://192.168.100.132/phpMyAdmin/webapp (CODE:200|SIZE:6903)

--- Entering directory: http://192.168.100.132/test/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

```

Gambar 14, Kali menjalankan dirb pada protokol HTTP Metasploitable 2

```

--- Entering directory: http://192.168.100.132/twiki/ ---
=> DIRECTORY: http://192.168.100.132/twiki/bin/
+ http://192.168.100.132/twiki/data (CODE:403|SIZE:298)
+ http://192.168.100.132/twiki/index (CODE:200|SIZE:782)
+ http://192.168.100.132/twiki/index.html (CODE:200|SIZE:782)
=> DIRECTORY: http://192.168.100.132/twiki/lib/
+ http://192.168.100.132/twiki/license (CODE:200|SIZE:19440)
=> DIRECTORY: http://192.168.100.132/twiki/pub/
+ http://192.168.100.132/twiki/readme (CODE:200|SIZE:4334)
+ http://192.168.100.132/twiki/templates (CODE:403|SIZE:303)

--- Entering directory: http://192.168.100.132/phpMyAdmin/contrib/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/js/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/lang/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/libraries/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/scripts/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/setup/ ---
+ http://192.168.100.132/phpMyAdmin/setup/config (CODE:303|SIZE:1370)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/frames/
+ http://192.168.100.132/phpMyAdmin/setup/index (CODE:200|SIZE:8619)
+ http://192.168.100.132/phpMyAdmin/setup/index.php (CODE:200|SIZE:8627)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/lib/
+ http://192.168.100.132/phpMyAdmin/setup/scripts (CODE:200|SIZE:21967)
+ http://192.168.100.132/phpMyAdmin/setup/styles (CODE:200|SIZE:6218)

--- Entering directory: http://192.168.100.132/phpMyAdmin/test/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

```

Gambar 15, Kali menjalankan dirb pada protokol HTTP Metasploitable 2

```
—— Entering directory: http://192.168.100.132/phpMyAdmin/themes/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—— Entering directory: http://192.168.100.132/twiki/bin/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—— Entering directory: http://192.168.100.132/twiki/lib/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—— Entering directory: http://192.168.100.132/twiki/pub/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—— Entering directory: http://192.168.100.132/phpMyAdmin/setup/frames/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—— Entering directory: http://192.168.100.132/phpMyAdmin/setup/lib/ ——  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—————  
END_TIME: Sat Jul 25 09:45:26 2026  
DOWNLOADED: 18448 - FOUND: 42
```

Gambar 16, Kali menjalankan dirb pada protokol HTTP Metasploitable 2